

European Option Convergence Study

Neel Dev Sen

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Introduction

This convergence study is based on the 3 European Option pricers I have made in my other project called **Pricing Engines** (which you can find on my **GitHub**). The numerical methods that will be compared is the Binomial Tree, Trinomial Tree and the Finite Differences (using the Crank-Nicolson scheme)

1 Theoretical and Numerical Framework

This section will mainly just be stating the Time and Space complexities, the complete derivations of these are in this **document**.

1.1 Time Complexities

The time complexity of a Binomial Tree with N steps is:

$$\boxed{\mathcal{O}(N^2)} \quad (1.1)$$

The time complexity of a Trinomial Tree with N steps is also:

$$\boxed{\mathcal{O}(N^2)} \quad (1.2)$$

Finally the time complexity of the Finite Differences with a mesh of M space nodes and N time nodes is:

$$\boxed{\mathcal{O}(M \times N)} \quad (1.3)$$

Since I used the Thomas algorithm it makes it faster instead of $\mathcal{O}(M^3 \times N)$ In this experiment to keep this all similar so it is easier to compare, I define $M = N$ to make a square mesh such that the time complexity of Finite Differences is also:

$$\boxed{\mathcal{O}(N^2)} \quad (1.4)$$

All of these time complexities will be useful when I show the runtime-convergence graphs of these numerical methods

1.2 Space Complexities

The total number of bytes allocated in a Binomial Tree with N steps is: $(2N - 1) * \text{size of type}$

Hence the space complexity of the Binomial Tree is:

$$\boxed{\mathcal{O}(N)} \quad (1.5)$$

The total number of bytes allocated in a Binomial Tree with N steps is: $(4N) * \text{size of type}$

Hence the space complexity of the Trinomial Tree is again:

$$\boxed{\mathcal{O}(N)} \quad (1.6)$$

Finally the time complexity of the Finite Differences with a mesh of N space nodes and N time nodes stores $(4N - 2) * \text{size of type bytes of memory}$ -

This gives us the space complexity of the Finite Differences to be:

$$\boxed{\mathcal{O}(N)} \quad (1.7)$$

1.3 Error Metrics

The **absolute error** is the main error metric that will be used to show convergence and it goes as shown.

$$\boxed{\text{Absolute Error} = |V_N - V_{CF}|} \quad (1.8)$$

where V_N is the value of the option given by the numerical method, and V_{CF} is the value of the option given by the closed form solution.

For the Arbitrage Error, I will just look at how far the options are from meeting the **Put-Call Parity** (the no-arbitrage condition) The Put Call Parity is as shown

$$C_t - P_t = S - Ke^{-r(T-t)} \quad (1.9)$$

The arbitrage error will be defined as:

$$\boxed{\text{Arbitrage Error} = |C_t - P_t - S + Ke^{-r(T-t)}|} \quad (1.10)$$

1.4 Data

The table below will be the data that is used in the convergence study.

Parameter	Value
Spot Price S_0	120.0
Strike K	100.0
Risk-Free Rate r	0.04
Volatility σ	0.25
Maturity T	2.0
Domain Multiplier	4.0

Table 1: Parameters used for the numerical convergence study.

2 Results

2.1 Black-Scholes Call

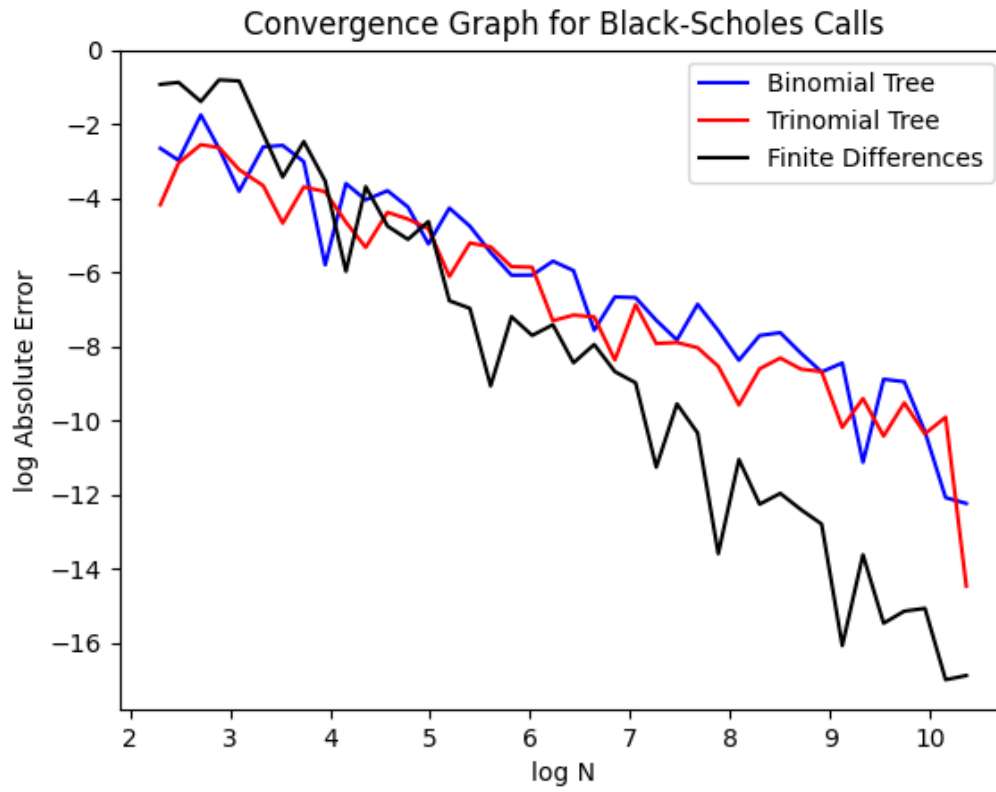


Figure 1: Convergence of the Black-Scholes call price.

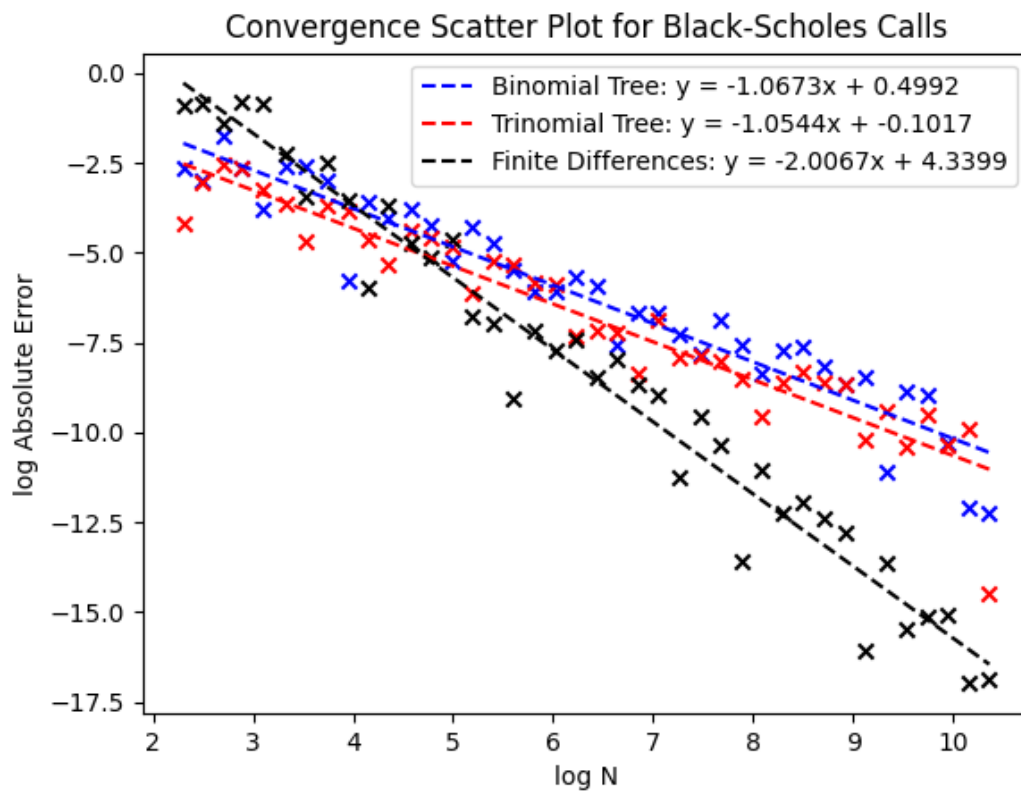


Figure 2: Convergence of the Black-Scholes call price.

2.2 Black-Scholes Put

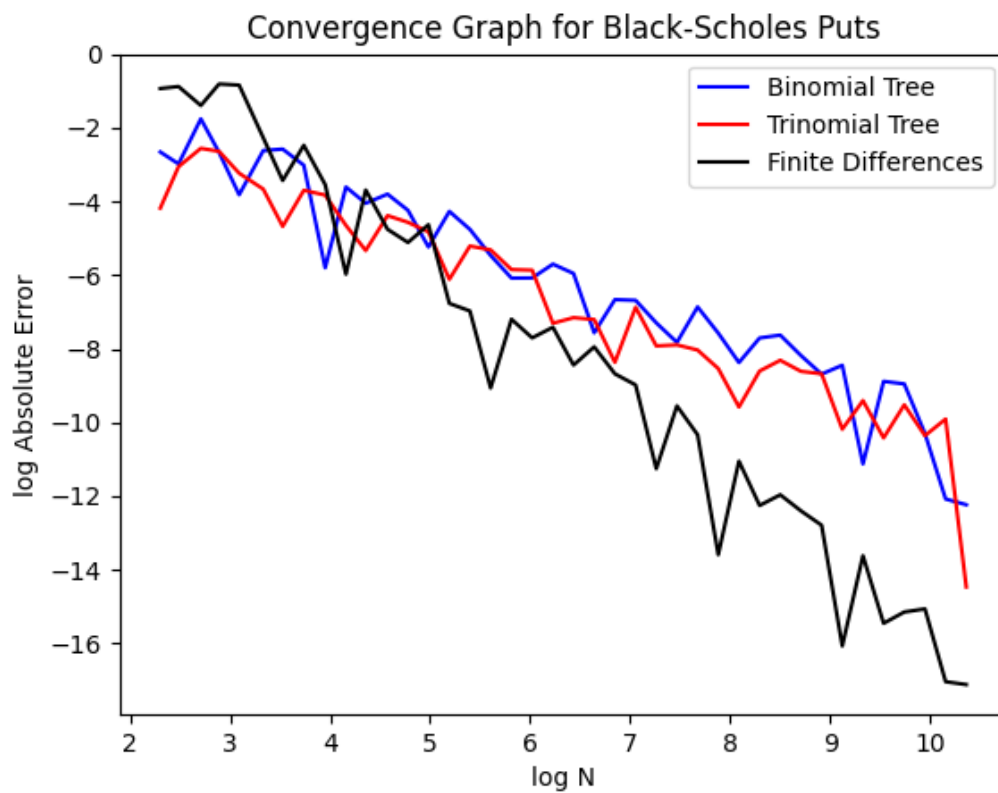


Figure 3: Convergence of the Black-Scholes put price.

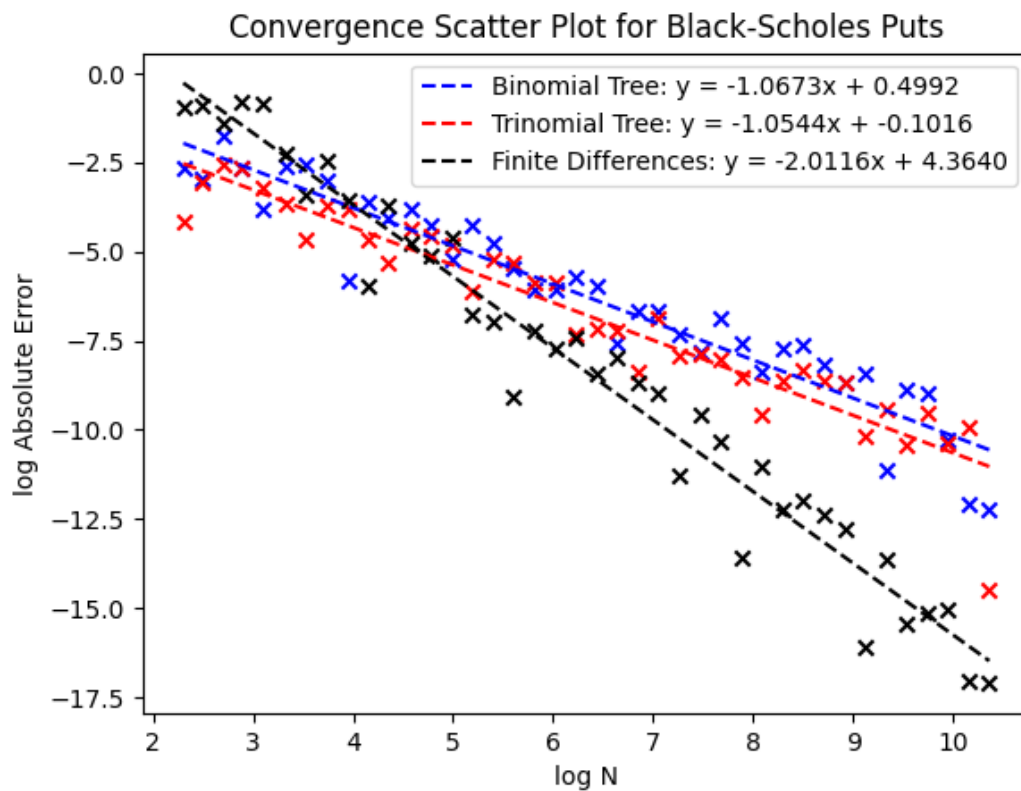


Figure 4: Convergence of the Black-Scholes call price.

2.3 Arbitrage Error

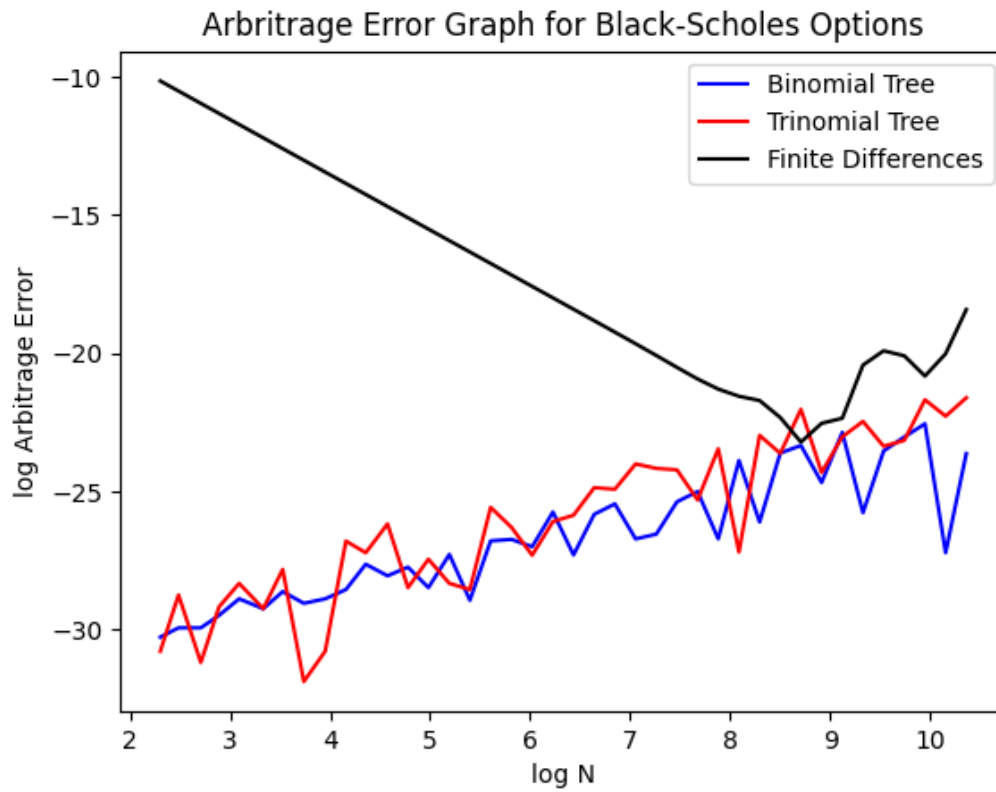


Figure 5: Effect of Steps on Arbitrage Error

2.4 Runtime

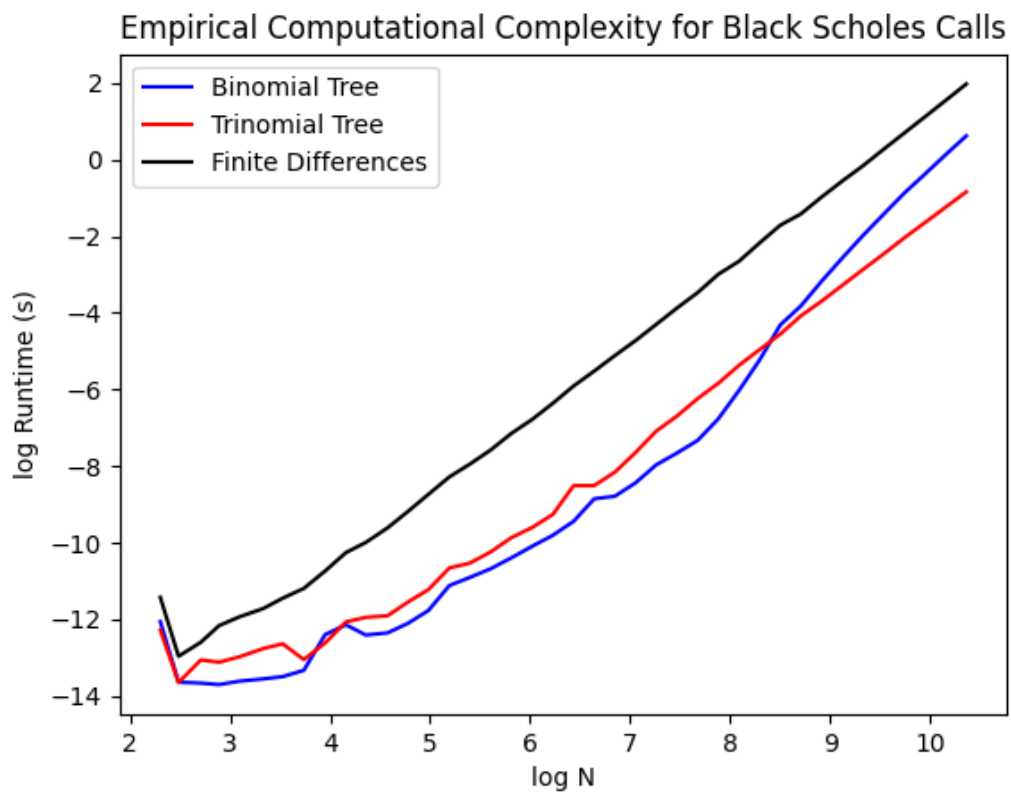


Figure 6: Effect of Steps on Arbitrage Error

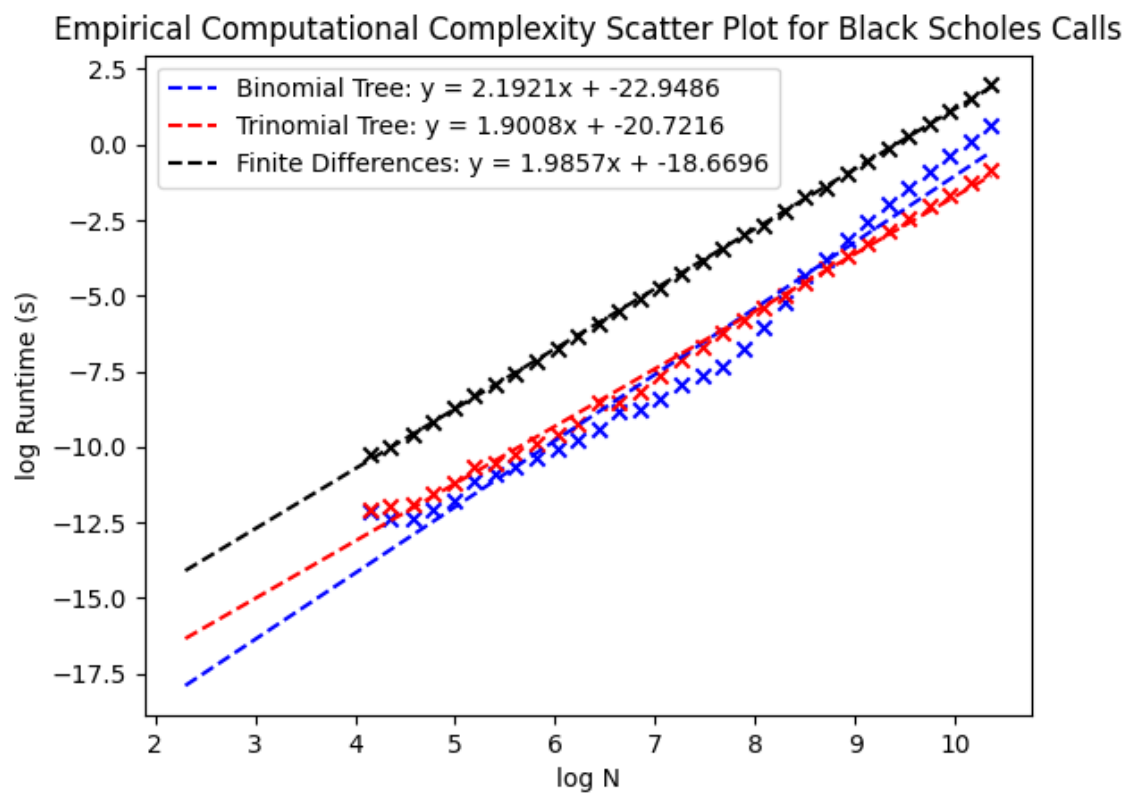


Figure 7: Effect of Steps on Arbitrage Error

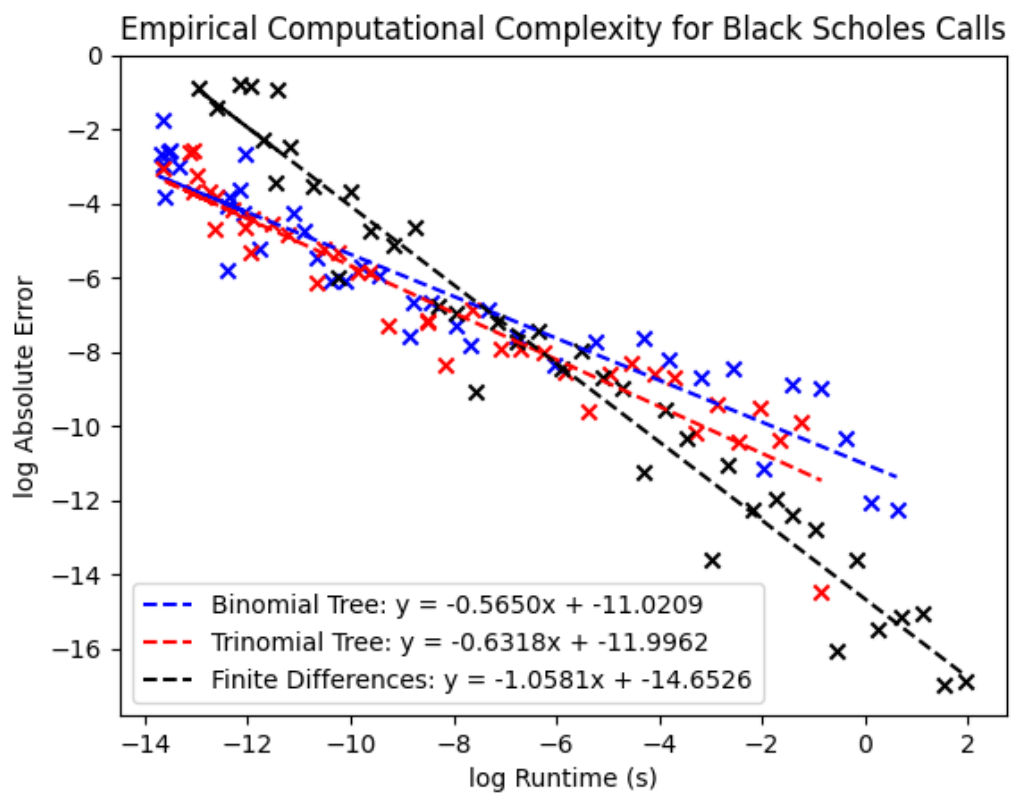


Figure 8: Effect of Steps on Arbitrage Error

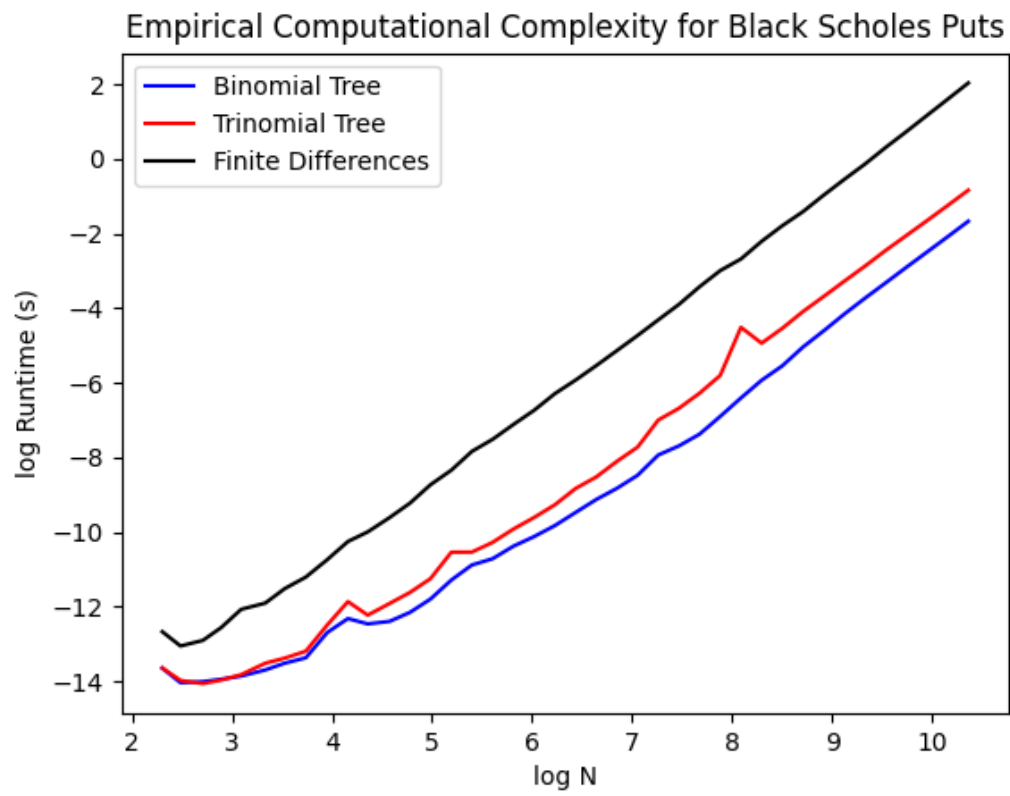


Figure 9: Effect of Steps on Arbitrage Error

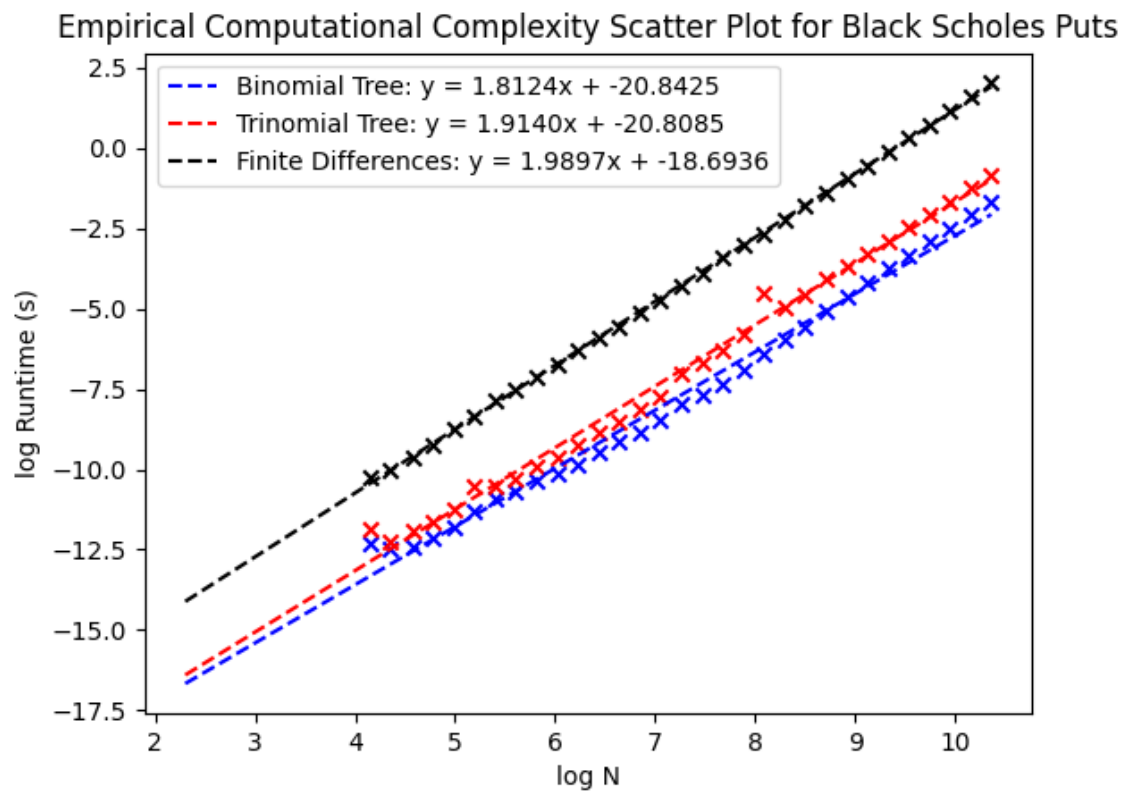


Figure 10: Effect of Steps on Arbitrage Error

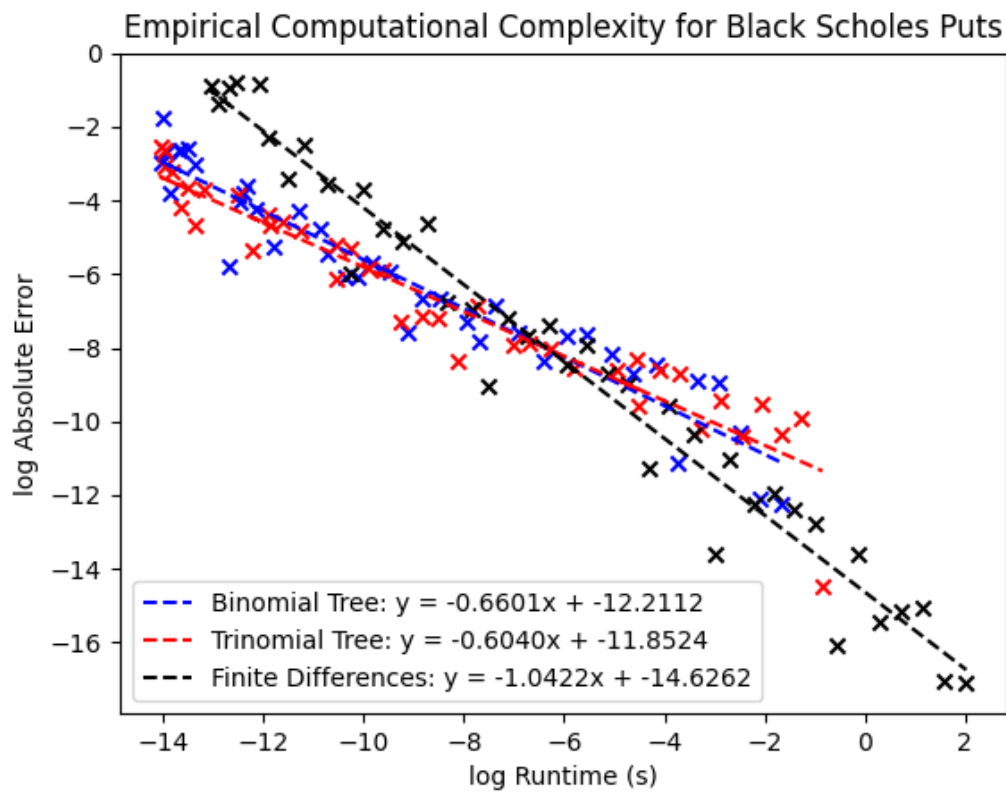


Figure 11: Effect of Steps on Arbitrage Error

2.5 Memory

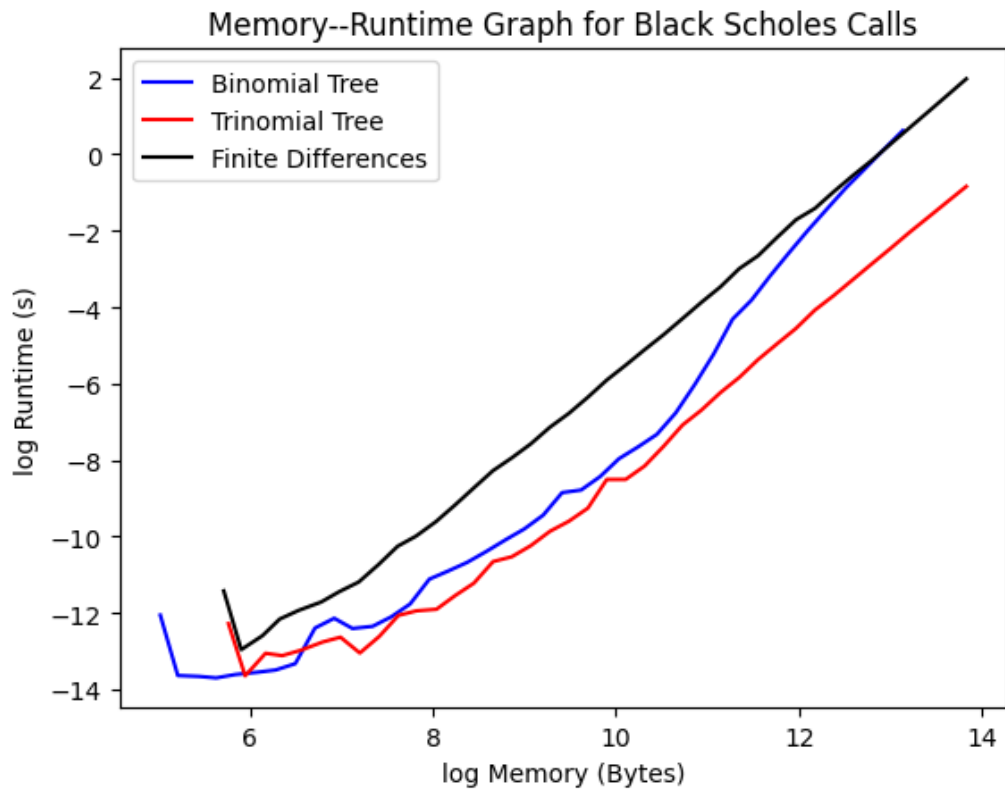


Figure 12: Effect of Steps on Arbitrage Error

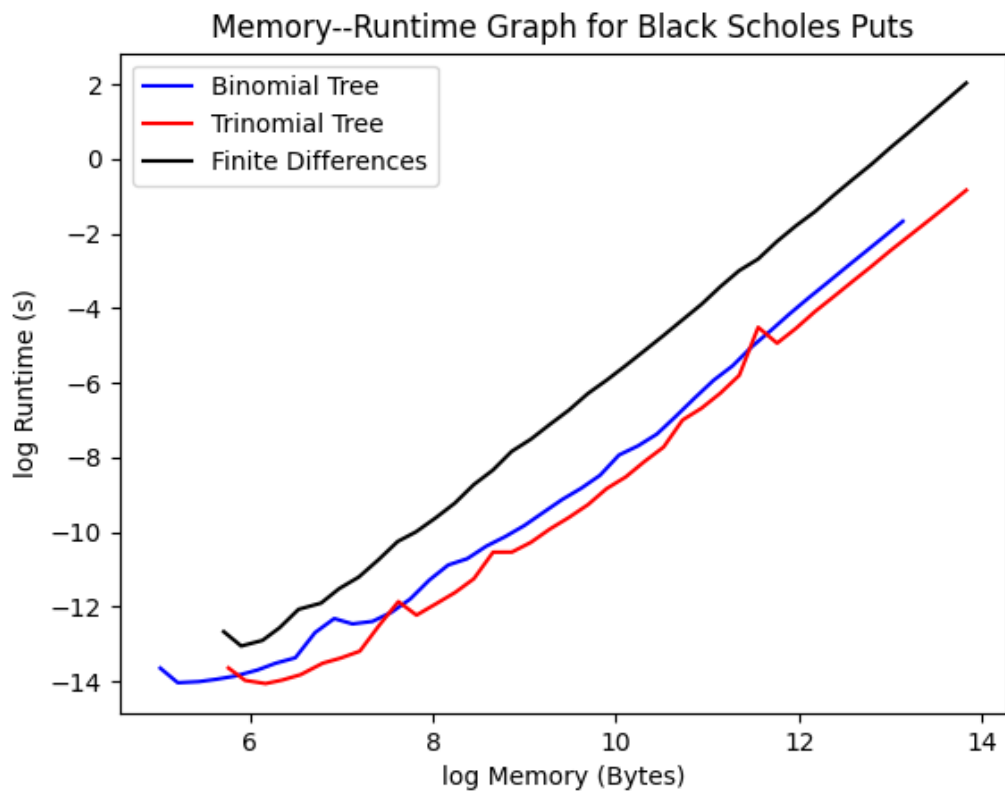


Figure 13: Effect of Steps on Arbitrage Error

3 Conclusion